BRAC UNIVERSITY

Inspiring Excellence

CSE471: System Analysis and Design
Assignment on Functional Requirements
Proposed Project Title: CitizenDesk

Group No: 4, CSE471 Lab Section: 9, [Fall] [2026]		
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Project Overview

Government services are scattered across different websites and offices, and each has its own confusing rules, fees, and forms. Because of this, people waste days dealing with paperwork, standing in long lines, or paying extra money to middlemen for help. CitizenDesk is a single, easy-to-use website that brings everything together. The system operates as a unified guide addressing four critical questions: What do I need?, What will it cost?, Where do I go?, and What happens next?

Tech Stack:

- Language: Typescript , Javascript
- Framework: Next.js
- Styling: TailwindCSS
- Database: PostgreSQL
- ORM: Prisma
- Deployment: Railway
- External APIs:

  Geoapify API: For interactive office geocoding, boundary mapping, and route directions

  Resend API : For automated deadline alerts

  Stripe API : For processing government fee payments.

  Gemini API: For automated local dialect translation

User Roles

1. **Citizen**: Uses the services. Can search and browse without an account, and once registered can save progress, use the checklist and fee calculator, keep documents in the vault, set reminders, track applications, pay fees, and file complaints.
2. **Government Employee**: Manages their own department's services and offices. Proposes changes to steps, documents, fees, and timelines, which go live after admin approval. Also keeps office details current, updates application stages, responds to complaints routed to their office, and confirms crowd-reported wait times. Cannot see citizens' vault documents or other departments' data.
3. **Admin**: Runs the platform. Approves employee registrations and employee-submitted changes, manages categories, life-event bundles, and the scam and official-domain watchlist, moderates crowd-reported data, manages users, and monitors payments and activity.

4. **Universal Service Search & Category Browser:** Users can type plain-language queries (e.g., "renew passport" or "birth certificate") into a universal search bar or browse via structured categories (identity, transport, tax, land, business, health, education, utilities, legal).
5. **Users can create an account using basic details** (name, email, phone number, password) and log in securely to access their dashboard, saved applications, and encrypted document vault.
6. **Admin can change Role & Access:** An authorized administrator signs in to a dedicated panel to verify official source links, moderate crowd-reported office wait times and spam, and update service guides, fees, and deadlines.

Functional Requirements

SL	Common Workflows
1	**Authentication, Access Control & User Profiles:** Multi-role registration, SSO/OAuth, secure login/logout, and profile attribute management (residence, contact details, user tier).
2	**Administrative Verification & Moderation Console:** Role-based administrative dashboard to inspect service records, review flagged community submissions, manage official domain lists, and audit "last verified" source timestamps.

Module 1	
Member	Feature Description
1	**Service Directory & Search :** A multi-category catalog covering identity, transport, tax, land, business, health, education, utilities, and legal sectors. Users can perform plain-language keyword search or browse by category. Each entry displays official source links, summary guidelines, and verification timestamps.

2	**Step-by-Step Walkthroughs:** An interactive guide that breaks down each administrative procedure into numbered steps distinguishing online workflows from in-person visit requirements. Users can tick interactive progress checkboxes to maintain state as they complete each phase.
3	**Personalized Document Requirements Checklist**: A rule-based questionnaire that evaluates user parameters (e.g., fresh application vs. renewal, age bracket, legal marital status) and dynamically generates an exact document checklist with specific requirements (e.g., "attested copy required", "2 passport photos").
4	**Fee & Timeline Calculator :** A calculation module that computes exact government charges based on citizen inputs (standard vs. express processing, age exemptions, penalty periods). The tool provides an itemized price sheet and estimated timeline to eliminate middleman overcharging.

Module 2	
Member	Feature Description
1	**Interactive Office Locator & Routing System:** A geospatial discovery interface utilizing the Geoapify API. Enables users to locate the nearest official ministry branch or regional office handling a specific service, view operational hours/holidays, and get turn-by-turn navigation.
2	**Rescheduling Appointments:** Citizens can reschedule their appointments
3	**Client-Side Encrypted Document Vault:** A private storage repository for citizens to safely upload, manage, and download identity documents and receipts. Uses AES-256 client-side encryption and strict access control to protect sensitive civil data.

4	**Government fee payments**: Users can pay official fees securely through integrated local payment APIs (such as bKash or digital banking gateways), ensuring instant payment confirmation, automated digital receipts, and a secure transaction audit trail using Stripe API.

Module 3	
Member	Feature Description
1	**Form Guide:** Provides annotated sample forms explaining each field and common mistakes, plus downloadable and printable blank forms.
1	**Plain-Language Translator :** Translates legal and civic jargon into plain phrasing in regional languages using **Gemini API**.
2	**Inter-Agency Prerequisite & Bottleneck Analyzer:**Rule-Based Prerequisite system, Uses an interactive logic tree where citizens answer 4–6 text-based questions about their current civil status.If a prerequisite is missing or outdated, the system dynamically recalculates their path, generates the exact resolution order, and tells them which sub-service must be completed first to unblock the target application.
2	**Scam Shield & Rights/Complaint Center:** Warns users about fake government websites, unofficial agents, and bribe demands using an integrated official domain link verifier. Generates formal complaint letters routed to the correct grievance office.
3	**Smart Digital Token & Queue Stage Tracker:**A real-time virtual token and stage-tracking system that monitors citizen task progress at physical

	offices and gives both citizens and desk officers live visibility over queues and processing speeds.
3	**FAQ and Help Center:** A searchable knowledge base that answers common questions about using CitizenDesk and about government services.
4	**Automated Deadline Reminder**: A recurring scheduler that tracks civil expiration deadlines (vehicle fitness certificates, trade licenses, passport renewals). Integrates with the Resend API to dispatch push notifications, text alerts, and email notifications before documents expire.
4	**Personalized Civic Activity Dashboard**: A single-pane-of-glass home console for authenticated citizens that consolidates active application tracking records, upcoming renewal deadlines, and saved document checklists.